Jacob Luttrull

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Education

- **Western Governors University** — Online
  Bachelor of Science in Computer Science — *Aug. 2023 – Aug. 2025*

Projects

- **Gator: RSS Feed Aggregator** — github.com/jacobrluttrull/gator
  Go, PostgreSQL, sqlc, goose — *July 2026*
  - **Architecture**: Built a Go RSS aggregator with 17 CLI commands and 13 REST endpoints sharing 1 PostgreSQL database
  - **Security**: Secured 11 authenticated endpoints with bcrypt password hashing and SHA-256-hashed API keys, isolating the trusted CLI from the network API
  - **Ingestion**: Automated feed fetching with a timed loop that polls the least-recently-updated feed first, running beside the API in 1 process
  - **Testing**: Verified behavior with integration tests and a curl-based smoke-test script covering the full API flow
- **Portfolio Website** — jacobrluttrull.com
  Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, pytest, Railway — *May 2026*
  - **Deployment**: Deployed a full-stack FastAPI and PostgreSQL site to Railway with Alembic-managed migrations and a JWT-protected admin panel
  - **Security**: Protected the public contact form with 4 layered defenses: XSS sanitization, CSRF tokens, Cloudflare Turnstile CAPTCHA, and rate limiting
  - **Reliability**: Hardened production with 3 security-header middlewares (CSP, HSTS, X-Frame-Options) and verified every public route and validator with a 71-test pytest suite
- **Oklahoma Sooners Dashboard** — github.com/jacobrluttrull/oklahoma_sooners_dashboard
  Python, Django, SQLite, Bootstrap, REST API — *Oct. 2025 – Nov. 2025*
  - **Data Model**: Modeled a normalized 5-table schema (Teams, Games, Players, team and player stats) serving records, AP rankings, leaders, schedule, and box scores from the CFBD API
  - **Sync**: Eliminated redundant API calls with idempotent schedule sync keyed by CFBD game ID and freshness-based caching
  - **Automation**: Reduced a full team and schedule refresh to 1 CLI command with a custom Django management command

Programming Skills

- **Languages**: Python, Go, SQL, Java, C, C++, HTML/CSS
- **Technologies**: FastAPI, Django, PostgreSQL, SQLite, SQLAlchemy, Alembic, sqlc, Docker, Git, Linux, pytest, Railway
- **Familiar**: Spring Boot, Svelte

Coursework & Certifications

- **Relevant Coursework**: Data Structures & Algorithms, Advanced Data Management (SQL), Boot.dev Back-end Developer Path (in progress: Go, Python, SQL, memory management)
- **Certifications**: ITIL Foundation (PeopleCert, expires May 2028), Linux Essentials (Linux Professional Institute)